18.336 FAST METHODS FOR PDES: FINAL REPORT
HIERARCHICAL POINCARÉ–STEKLOV SOLVERS FOR 2D LINEAR ELASTICITY

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Abstract. The hierarchical Poincaré–Steklov (HPS) solver is applied to the two-dimensional (plane-strain) isotropic linear elasticity. The Navier–Cauchy system of equations is split via an ADI iteration that treats the cross-derivative coupling term as a source, allowing the HPS build stage to be executed once per displacement component. Two benchmark problems are considered. The first is a smooth manufactured-solution problem for which spectral-exponential convergence in polynomial degree p is expected and observed. The second is a Mode I crack-tip problem whose exact solution, given by the Williams asymptotic expansion, carries a $\sqrt{r}$ corner singularity that degrades convergence to algebraic $\mathcal{O}(p^{-1/2})$. Convergence studies are reported for different levels of quadtree refinement and polynomial orders, and results are compared with analytical solutions.

Key words. hierarchical Poincaré–Steklov, linear elasticity, crack-tip singularity, JAX

1. Introduction. The field of solid mechanics has been historically dominated by the finite element method, which has been the workhorse for simulating the deformation and failure of materials and structures. Classical FEM methods achieve algebraic convergence with mesh refinement, but reaching high accuracy requires either a very fine mesh (h-refinement) or high polynomial degrees (p-refinement). High-order spectral methods can attain *exponential* convergence for smooth solutions, achieving high accuracy with fewer degrees of freedom. However, they are often considered less flexible than FEM for complex geometries and boundary conditions.

Among high-order approaches, *fast direct solvers* are particularly attractive for problems, where the PDE must be solved many times for different data. Once the operator is factored (the expensive "build" stage), each new right-hand side or boundary datum can be applied in a rapid "solve" stage at the cost of a few matrix-vector products. The *hierarchical Poincaré–Steklov* (HPS) family of algorithms was introduced by Martinsson [8] and extended to compression schemes, iterative variants and GPU-accelerated implementations in a series of works [4, 5, 9]. The scheme operates on variable-coefficient elliptic PDEs.

The HPS framework, as implemented in the open-source JAX library `jaxhps` [9], is applied here to *two-dimensional plane-strain linear elasticity* governed by the Navier Cauchy equations. Because these equations couple the two displacement components through a mixed cross-derivative term, the coupled system cannot be solved directly using the scalar HPS solver. At this point, one might consider reformulating the system as a 4th-order PDE for a single potential (Airy stress function), but this would lead to more complicated boundary conditions and operator. Instead, an *Alternating Direction Implicit* (ADI) splitting is adopted that treats the cross-derivative term as an explicit source and alternates between decoupled scalar problems for u_x and u_y. The differential operator is the same at every ADI iteration, so the HPS build stage is executed only once per component; subsequent iterations perform only the cheap downward pass.

Two benchmarks illustrate the method's strengths and limitations:

1. **Smooth manufactured solution.** The exact solution $u_x = \sin(\pi x)\sin(\pi y)$, $u_y = \cos(\pi x)\cos(\pi y)$ is analytic, so exponential convergence in p is expected and observed.
2. **Mode I crack-tip benchmark.** The Williams asymptotic expansion, which

has a $\sqrt{r}$ corner singularity, limits convergence to algebraic $\mathcal{O}(p^{-1/2})$ regardless of whether HPS or FEM is used. The purpose is to show the fundamental limitation of polynomial approximation near singularities.

2. Background and Related Work.

High-order direct solvers. The idea of combining composite spectral collocation with nested dissection was first proposed by Martinsson [8], who built a direct solver for variable-coefficient elliptic PDEs by merging leaf-level Dirichlet-to-Neumann operators up a quadtree. Gillman and Martinsson [4] introduced a compression scheme that exploits the low-rank structure of off-diagonal blocks in the DtN matrices, reducing complexity to $\mathcal{O}(N)$ for N degrees of freedom. Gillman, Young, and Martinsson [5] replaced the DtN merge with an impedance-to-impedance (ItI) merge to improve numerical stability for high-frequency Helmholtz problems. Extensions to three dimensions include Hao and Martinsson [6] and Lucero Lorca et al. [7]. Melia et al. [9] ported the method to JAX [2], enabling GPU acceleration and automatic differentiation and explored strategies for efficient implementation.

hp-FEM and singularities. The finite element method has been extensively studied for problems with reentrant corners and singularities. The Sobolev regularity of the Williams solution is $H^{3/2-\epsilon}$, which implies that the best approximation error in the H^1 seminorm satisfies $\|u - u_h\|_{H^1} = \mathcal{O}(h^{1/2})$ for uniform h-refinement, independent of the polynomial order p [3]. This rate $h^{1/2}$ is confirmed for a standard Continuous Galerkin discretization in Figure 10 (section B). The same barrier is expected to apply to any scheme that uses global polynomial approximation including HPS. Remedies include geometric (h-) refinement toward the singularity (Babuška and Rheinboldt, 1978 [1]) or enrichment with singular basis functions (the eXtended FEM approach of Moës et al., 1999 [10]). This report does not attempt these remedies, which could be explored in future work.

Linear fracture mechanics. A crack tip can be loaded in three independent modes, shown in Figure 1: Mode I (opening), Mode II (in-plane shear), and Mode III (out-of-plane shear) [11]. This report focuses exclusively on Mode I, the most common in structural applications. The near-tip displacement field is given by the Williams asymptotic expansion, and the amplitude of the singularity is characterised by the stress-intensity factor K_I. The small-scale yielding assumption is made that the plastic zone remains confined to the crack tip region (Figure 5), so the linear-elastic model applies almost everywhere in Ω.

3. Governing Equations: 2D Linear Elasticity. The domain $\Omega \subset \mathbb{R}^2$ is occupied by a homogeneous, isotropic, linearly elastic body under plane-strain conditions. The displacement field $\mathbf{u} = (u_x, u_y)^T$ satisfies the Navier–Cauchy equations

$$\begin{pmatrix} (\lambda+2\mu)\partial_{xx} + \mu\partial_{yy} & (\lambda+\mu)\partial_{xy} \\ (\lambda+\mu)\partial_{xy} & \mu\partial_{xx} + (\lambda+2\mu)\partial_{yy} \end{pmatrix} \begin{pmatrix} u_x \\ u_y \end{pmatrix} = -\begin{pmatrix} f_x \\ f_y \end{pmatrix} \tag{3.1}$$

in Ω, subject to Dirichlet boundary conditions $\mathbf{u} = \mathbf{g}$ on $\partial\Omega$. Here λ and μ are the Lamé constants (material parameters) and $\mathbf{f} = (f_x, f_y)^T$ is the body force per unit volume.

The stress tensor and strain tensor for an isotropic material in plane strain are

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \tfrac{1}{2}\big(\nabla\mathbf{u} + (\nabla\mathbf{u})^T\big), \tag{3.2}$$

$$\boldsymbol{\sigma}(\mathbf{u}) = \lambda\,(\operatorname{tr}\boldsymbol{\varepsilon})\,\mathbf{I} + 2\mu\,\boldsymbol{\varepsilon}, \tag{3.3}$$

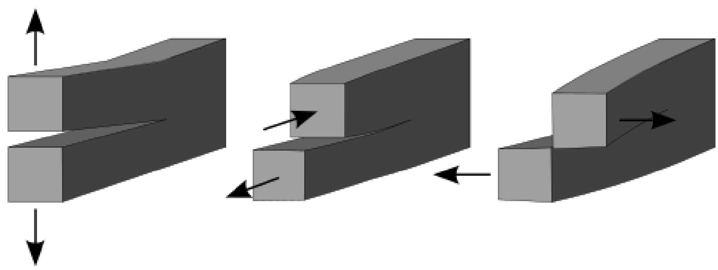

Fig. 1: The three independent fracture modes at a crack tip. Mode I (opening): symmetric tensile loading perpendicular to the crack plane. Mode II (in-plane shear): sliding of the crack faces parallel to the crack plane and perpendicular to the crack front. Mode III (out-of-plane shear): tearing, with crack-face displacements parallel to the crack front. The general stress state is a superposition of all three, each weighted by its stress-intensity factor K_I, K_{II}, K_{III}.

The von Mises equivalent stress is

$$\sigma_{\mathrm{vm}} = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\sigma_{xy}^2}. \tag{3.4}$$

The Poisson ratio in plane strain and the Kolosov constant are

$$\nu = \frac{\lambda}{2(\lambda+\mu)}, \qquad \kappa = 3 - 4\nu. \tag{3.5}$$

4. The Hierarchical Poincaré–Steklov (HPS) Algorithm. A short description of the HPS algorithm [8, 9] is given below for the scalar second-order elliptic boundary-value problem

$$\mathcal{L}u = f \quad \text{in } \Omega, \qquad u = g \quad \text{on } \partial\Omega. \tag{4.1}$$

The method proceeds in three stages: local solve, merge (upward pass) and downward pass.

4.1. Poincaré–Steklov operators. A *Poincaré–Steklov operator* $T : g \mapsto h$ maps one type of boundary data to another for the equation $\mathcal{L}u = f$ on a subdomain. The most common instance is the *Dirichlet-to-Neumann* (DtN) map: given Dirichlet data $g = u|_{\partial\Omega_j}$, it returns the outward normal derivative $h = \partial_n u|_{\partial\Omega_j}$. The key property exploited by the HPS algorithm is that these operators can be *merged*: the DtN operator for a composite domain $\Omega = \Omega_a \cup \Omega_b$ can be expressed purely in terms of the DtN operators of Ω_a and Ω_b via a Schur complement, without ever solving the PDE on the full domain.

Because $\mathcal{L}$ is linear, the solution decomposes as $u = v + w$, where v solves $\mathcal{L}v = f$ with zero Dirichlet data (particular solution) and w solves $\mathcal{L}w = 0$ with Dirichlet data g (homogeneous solution). The operator T is therefore the DtN map for the homogeneous problem and both $T^{(i)}$ and the associated particular-solution flux $\mathbf{h}^{(i)} = \partial_n v^{(i)}|_{\partial\Omega_i}$ can be precomputed before the boundary data g is known.

4.2. Discretization. The domain Ω is recursively partitioned into a quadtree of depth L, yielding $n_{\text{leaves}} = 4^L$ leaf patches. Figure 2 shows the two key grids for a simple two-dimensional example with $L = 1$ and $p = 8$. Each leaf is discretized on a $p \times p$ tensor-product Chebyshev–Lobatto grid (panel (a)), giving p^2 interior degrees of freedom per leaf and a total of $N = 4^L p^2$ interior unknowns. Leaf boundaries are discretized on q-point Gauss–Legendre grids with $q = p - 2$, following [5]. Panel (b) distinguishes the *exterior* boundary points (blue circles, on $\partial\Omega$) from the *interior interface* points (red crosses, shared between adjacent patches $\Omega_a, \Omega_b, \Omega_c, \Omega_d$); this distinction is central to the merge step.

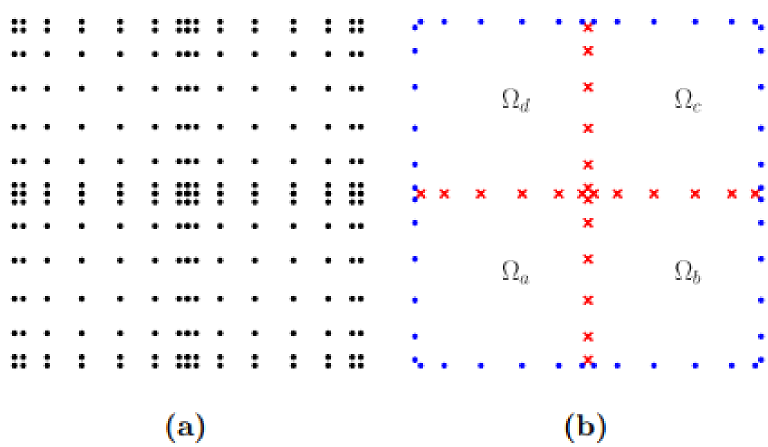

Fig. 2: High-order spectral collocation for a 2D problem with $p = 8$, $L = 1$. **(a)** Chebyshev–Lobatto interior points on all four leaves. **(b)** Gauss–Legendre boundary points: blue circles are exterior points on $\partial\Omega$; red crosses are interior interface points shared by adjacent patches Ω_a, Ω_b, Ω_c, Ω_d. (Figure adapted from [9].)

4.3. Local solve stage. On each leaf i, the spectral differentiation matrices are assembled into the discrete operator $\mathbf{L}^{(i)} \in \mathbb{R}^{p^2 \times p^2}$. A *boundary-bordering* technique augments this system so that Dirichlet data can be enforced along the leaf boundary while the PDE is collocated at all interior Chebyshev nodes. The bordered system is inverted once to extract:

- $\mathbf{Y}^{(i)} \in \mathbb{R}^{p^2 \times q_{\text{bdry}}}$: the *solution operator*, mapping any leaf boundary data to the corresponding homogeneous solution at interior Chebyshev nodes.
- $\mathbf{v}^{(i)} \in \mathbb{R}^{p^2}$: the particular solution at interior nodes.

Applying a fixed interpolation–differentiation operator to $\mathbf{Y}^{(i)}$ and $\mathbf{v}^{(i)}$ yields the leaf-level DtN matrix $\mathbf{T}^{(i)}$ and the particular-solution outgoing data $\mathbf{h}^{(i)}$. This stage is *embarrassingly parallel* across all 4^L leaves, and its cost is $\mathcal{O}(p^6 n_{\text{leaves}})$ (dominated by the $p^2 \times p^2$ matrix inversion on each leaf).

4.4. Merge (upward pass) stage. Starting from the leaves, the algorithm merges groups of four sibling nodes into their parent until the root is reached. Let j be a parent node with children $\{a, b, c, d\}$. Define index sets $\mathcal{E}$ (exterior boundary points of j, blue in Figure 2(b)) and $\mathcal{I}$ (interior interface points, red). Using block notation, the DtN matrices of the children are assembled into

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \mathbf{g}_{\mathcal{E}} \\ \mathbf{g}_{\mathcal{I}} \end{bmatrix} = \begin{bmatrix} \mathbf{u}_{\mathcal{E}} - \mathbf{h}_{\mathcal{E}}^{(\text{child})} \\ -\mathbf{h}_{\mathcal{I}}^{(\text{child})} \end{bmatrix}, \tag{4.2}$$

where A, B, C, D are assembled blockwise from $\{\mathbf{T}^{(a)}, \mathbf{T}^{(b)}, \mathbf{T}^{(c)}, \mathbf{T}^{(d)}\}$ and the continuity conditions across merge interfaces. Since $\mathbf{g}_{\mathcal{E}}$ (the exterior boundary data of node j) is not yet known, $\mathbf{g}_{\mathcal{I}}$ is eliminated via a Schur complement on D:

$$\mathbf{T}^{(j)} = A - BD^{-1}C, \tag{4.3}$$

$$\mathbf{h}^{(j)} = \mathbf{h}_{\mathcal{E}}^{(\text{child})} - BD^{-1}\mathbf{h}_{\mathcal{I}}^{(\text{child})}, \tag{4.4}$$

$$\mathbf{S}^{(j)} = -D^{-1}C, \tag{4.5}$$

$$\tilde{\mathbf{g}}^{(j)} = -D^{-1}\mathbf{h}_{\mathcal{I}}^{(\text{child})}. \tag{4.6}$$

Here $\mathbf{T}^{(j)}$ and $\mathbf{h}^{(j)}$ are passed to the next merge level; $\mathbf{S}^{(j)}$ (the *propagation operator*) and $\tilde{\mathbf{g}}^{(j)}$ (particular-solution data at the interior interfaces) are stored for the downward pass. The dominant cost at each level is inverting D, whose size grows as $\mathcal{O}(pq^{L-\ell})$ at level ℓ; the total merge cost is $\mathcal{O}(p^3 n_{\text{leaves}}^{3/2})$ in 2D.

4.5. Downward pass. Once the boundary data $\mathbf{g}$ on $\partial\Omega$ is available, the precomputed operators are applied in a single top-to-bottom sweep of the tree:

$$\mathbf{g}_{\mathcal{I}}^{(j)} = \mathbf{S}^{(j)}\mathbf{g}_{\mathcal{E}}^{(j)} + \tilde{\mathbf{g}}^{(j)}, \tag{4.7}$$

propagating boundary data from each node to the boundaries of its children. At the leaf level:

$$\mathbf{u}^{(i)} = \mathbf{Y}^{(i)}\mathbf{g}^{(i)} + \mathbf{v}^{(i)}. \tag{4.8}$$

The downward pass consists entirely of matrix-vector products with cost $\mathcal{O}(p^2 n_{\text{leaves}})$, and it can be re-executed in negligible time for new boundary data $\mathbf{g}$ or new sources f (after recomputing only $\mathbf{v}^{(i)}$ and $\tilde{\mathbf{g}}^{(j)}$).

4.6. Complexity summary. The local solve stage costs $\mathcal{O}(p^6 n_{\text{leaves}})$ and is embarrassingly parallel over leaves. The merge (upward pass) costs $\mathcal{O}(p^3 n_{\text{leaves}}^{3/2})$ with parallelism within each tree level. The downward pass costs $\mathcal{O}(p^2 n_{\text{leaves}})$. The build stage (local solve + merge) is performed *once* per operator. The downward pass is re-applied for each new right-hand side at a small cost, making HPS highly efficient for problems requiring many solves with the same operator.

5. ADI Splitting for Coupled Elasticity. The Navier–Cauchy system (3.1) couples u_x and u_y through the $(\lambda+\mu)\partial_{xy}$ off-diagonal terms, preventing direct use of the scalar HPS solver. The key observation is that if the coupling is treated as a known source, each component satisfies a decoupled scalar elliptic equation of the form (4.1). This motivates an alternating fixed-point iteration. At each step k, u_x is solved first with the cross-derivative term from the previous iterate $u_y^{(k-1)}$; then u_y is solved using the updated $u_x^{(k)}$:

$$\left[(\lambda+2\mu)\partial_{xx} + \mu\partial_{yy}\right] u_x^{(k)} = -f_x - (\lambda+\mu)\,\partial_{xy}\,u_y^{(k-1)}, \quad u_x^{(k)}\big|_{\partial\Omega} = g_x, \tag{5.1a}$$

$$\left[\mu\partial_{xx} + (\lambda+2\mu)\partial_{yy}\right] u_y^{(k)} = -f_y - (\lambda+\mu)\,\partial_{xy}\,u_x^{(k)}, \quad u_y^{(k)}\big|_{\partial\Omega} = g_y. \tag{5.1b}$$

Each subproblem is a scalar elliptic boundary value problem to which HPS applies directly.

Cost structure.. Since the operator in each subproblem of (5.1) is fixed across iterations, `build_solver` is called *once* per displacement component, storing the factored operators $\mathbf{Y}^{(i)}, \mathbf{S}^{(j)}, \mathbf{T}^{(j)}$. Each ADI iteration then calls only the inexpensive `solve` (downward pass), plus a batched spectral cross-derivative product at cost $\mathcal{O}(p^4 n_{\text{leaves}})$. The assumption is that the ADI iteration converges in a small number of steps, independent of the discretization size.

Convergence criterion.. Iterations continue until the ℓ^∞ displacement update falls below tol $= 10^{-14}$:

$$\max_i \left(\|u_x^{(k)} - u_x^{(k-1)}\|_\infty + \|u_y^{(k)} - u_y^{(k-1)}\|_\infty \right) < \text{tol}, \tag{5.2}$$

ensuring iteration errors remain well below the discretization error for all polynomial degrees tested.

6. Case 1: Smooth Manufactured Solution.

6.1. Convergence. The problem is solved on $\Omega = [-1,1]^2$ with $\lambda = \mu = 1$ and exact solution $u_x = \sin(\pi x)\sin(\pi y)$, $u_y = \cos(\pi x)\cos(\pi y)$, with body forces $f_x = 2\mu\pi^2 \sin(\pi x)\sin(\pi y)$ and $f_y = 2\mu\pi^2 \cos(\pi x)\cos(\pi y)$. The sweep covers $p \in \{3,4,6,8,12,16\}$ for $L \in \{2,3,4\}$.

Figure 3 shows the relative L^∞ error in u_x and u_y as a function of p on a semi-log scale. The curves demonstrate *spectral-exponential* convergence; errors reach double-precision ($\approx 10^{-13}$) at $p = 12$ for $L \geq 3$.

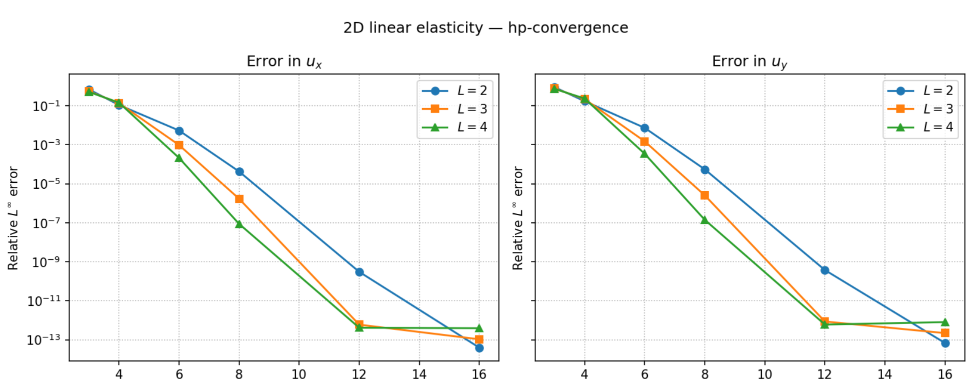

Fig. 3: Relative L^∞ error vs. polynomial degree p for the smooth manufactured solution on $[-1,1]^2$. Left: error in u_x; right: error in u_y. Each curve corresponds to a quadtree depth $L \in \{2,3,4\}$. Errors decay exponentially until double-precision saturation near 10^{-13}.

6.2. Solution Fields and ADI Iterations. Figure 4 shows displacement and pointwise absolute error at $L = 4$, $p = 16$; all errors are at or below 5×10^{-13}. For all (p, L) tested, the ADI solver converged in fewer than 10 iterations, essentially independent of L.

7. Case 2: Mode I Crack-Tip Singularity.

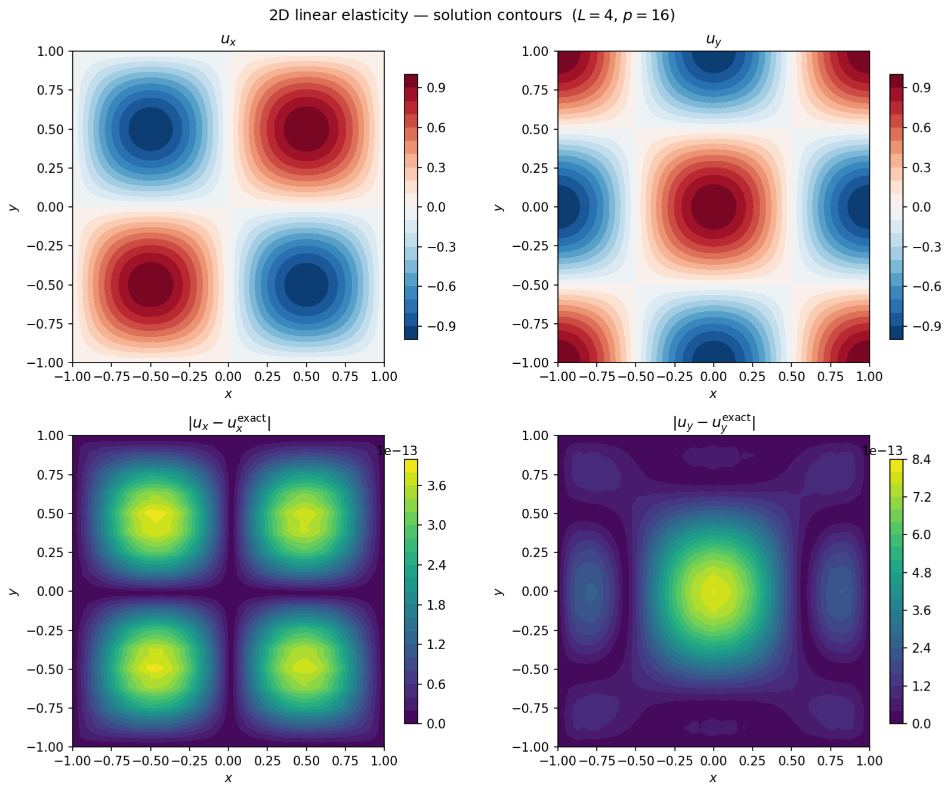

Fig. 4: Numerical solution (top) and pointwise absolute error (bottom) for the smooth manufactured problem at $L = 4$, $p = 16$. Top left: u_x; top right: u_y. Bottom left: $|u_x - u_x^{\text{exact}}|$; bottom right: $|u_y - u_y^{\text{exact}}|$. All errors are at or below 5×10^{-13}.

7.1. Problem Setup and Exact Solution. For this mode-I crack-tip problem, the domain is $\Omega = [0,1]^2$, with the crack tip at the corner $(0,0)$. The crack lies along the negative x-axis, entirely outside $[0,1]^2$, so the solution is single-valued in Ω; however, the $\sqrt{r}$ behaviour near the origin constitutes a corner singularity.

Figure 5 illustrates the classical plastic zone surrounding a Mode I crack tip. Inside this zone the material yields and the linear-elastic model breaks down. However, this region is assumed to be small in this problem, so the linear-elastic solution is valid almost everywhere in Ω.

The exact solution is the Williams Mode I asymptotic expansion [11]:

$$u_x = \frac{K_I}{2\mu}\sqrt{\frac{r}{2\pi}}\cos\frac{\theta}{2}\big(\kappa - \cos\theta\big), \tag{7.1}$$

$$u_y = \frac{K_I}{2\mu}\sqrt{\frac{r}{2\pi}}\sin\frac{\theta}{2}\big(\kappa - \cos\theta\big), \tag{7.2}$$

with $r = \sqrt{x^2 + y^2}$, $\theta = \operatorname{atan2}(y, x)$, Kolosov constant $\kappa = 3 - 4\nu$, and $\nu = \lambda/(2(\lambda+\mu))$ the plane-strain Poisson ratio. Parameters are set to $\lambda = \mu = K_I = 1$, giving $\nu = 1/4$ and $\kappa = 2$. The governing equations are (3.1) with $\mathbf{f} = \mathbf{0}$, and Dirichlet data on all

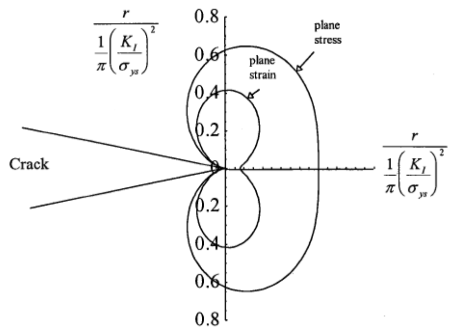

Fig. 5: Irwin plastic zone for a Mode I crack. The zone boundary for plane stress (outer curve) and plane strain (inner curve) are plotted in coordinates normalised by $r_p = \frac{1}{\pi}(K_I/\sigma_{ys})^2$.

four sides of $[0,1]^2$ is set exactly to (7.1) and (7.2).

7.2. Convergence Results. Because $\mathbf{u} \sim r^{1/2}$ near the origin (Sobolev regularity $H^{3/2-\epsilon}$), best-approximation theory predicts algebraic convergence [3]:

$$\|u - u_p\|_{L^\infty} = \mathcal{O}(p^{-1/2}), \tag{7.3}$$

independent of the mesh refinement level. This barrier is universal: standard FEM shows the same rate (see Figure 10 in section B).

Figure 6 shows the relative L^∞ errors for the crack-tip problem on a log-log scale, confirming the predicted algebraic rate $\mathcal{O}(p^{-1/2})$. The error at $p = 16$, $L = 4$ remains above 1%, several orders of magnitude above the machine-precision floor reached for the smooth problem. Finer meshes (larger L) consistently reduce errors at fixed p by shrinking the corner patch, but do not improve the algebraic convergence rate.

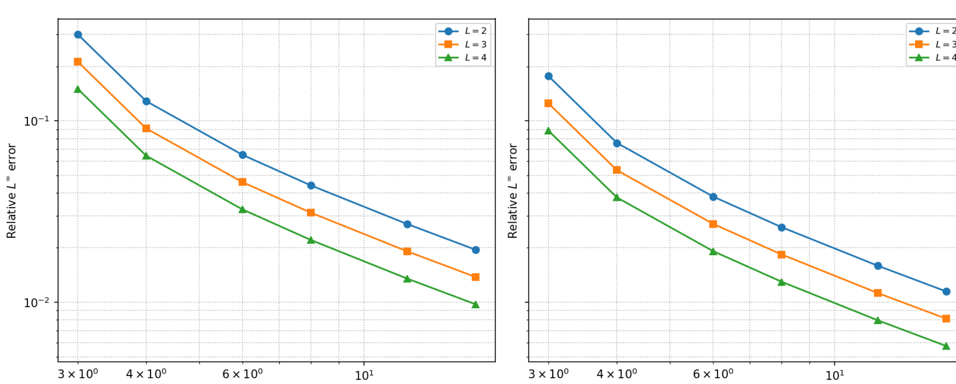

Fig. 6: Relative L^∞ error vs. polynomial degree p for the Mode I crack-tip problem (log-log scale). Left: error in u_x; right: error in u_y. Convergence is algebraic at rate $\mathcal{O}(p^{-1/2})$. Increasing L reduces errors at fixed p by shrinking the corner patch.

Solution fields.. Figure 7a shows the displacement components u_x and u_y at $L = 4$, $p = 16$; both fields radiate from the crack tip with the $\sqrt{r}$ growth of the Williams expansion. Figure 7b shows the corresponding Cauchy stress components and von Mises stress, all exhibiting the classical $1/\sqrt{r}$ singularity; qualitative features match textbook fracture mechanics results.

ADI iteration count.. With zero body force the right-hand side comes entirely from the cross-derivative coupling term. Figure 8 shows the ADI iteration counts: the solver converges in fewer than 20 iterations for all p and L tested, slightly more than the smooth case because the singular $\sqrt{r}$ solution yields larger cross-derivative values near the corner.

7.3. Computational Cost. Figure 9 shows wall-clock time (build + solve) for both problems as a function of p and L. Run times increase with both p and L, dominated by the build stage. Both problems show nearly identical timing profiles; the slightly larger ADI iteration count for the crack-tip case is negligible relative to the build time.

8. Conclusions. Results demonstrate that the HPS fast direct solver, combined with an ADI splitting to handle the cross-derivative coupling in the Navier–Cauchy equations, is a practical and efficient way to solve two-dimensional plane-strain linear elasticity problems.

The key findings are:

1. **Spectral-exponential convergence for smooth problems.** For the smooth manufactured solution, errors reach machine precision at moderate polynomial degrees ($p \approx 12$) regardless of the refinement level L, confirming the spectral accuracy of the composite Chebyshev collocation scheme.
2. **Algebraic convergence for singular problems.** For the Mode I crack-tip problem, convergence degrades to the theoretically predicted $\mathcal{O}(p^{-1/2})$ algebraic rate. Errors remain above 1% even at $p = 16$, highlighting the fundamental limitation that corner singularities impose on polynomial approximation.
3. **Efficient ADI splitting.** The ADI scheme converges in fewer than 10 iterations for the smooth problem and fewer than 20 for the singular problem, with the build stage executed only once per component.

Directions for future work include:

- *Adaptive h-refinement* concentrated near the singularity; `jaxhps` already supports adaptive quadtree refinement, making this extension straightforward.
- GPU acceleration using the JAX backend of `jaxhps` [9], which could enable much larger problem sizes and three-dimensional elasticity.
- The incompressible limit ($\nu \to 1/2$), where $\lambda \to \infty$ strengthens the cross-derivative coupling.
- Other singularity types (e.g. material interfaces) and their impact on convergence.

Acknowledgments. The numerical experiments in this report were carried out using `jaxhps` [9], an open-source JAX implementation of the hierarchical Poincaré–Steklov algorithm developed by Melia et al. The library is publicly available at https://github.com/meliao/jaxhps.

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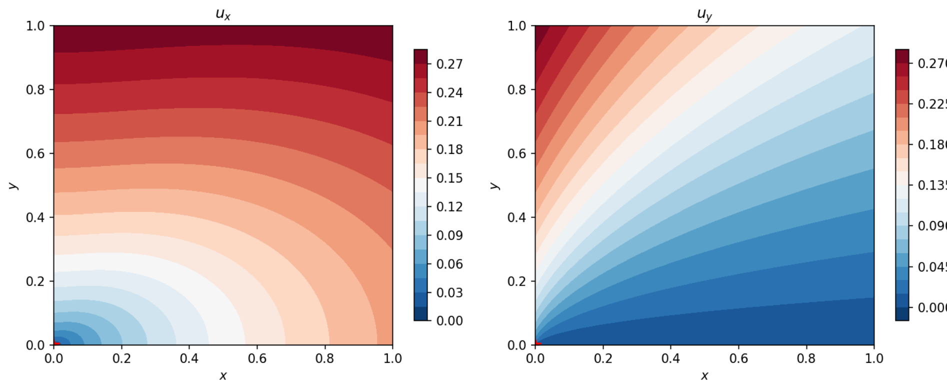
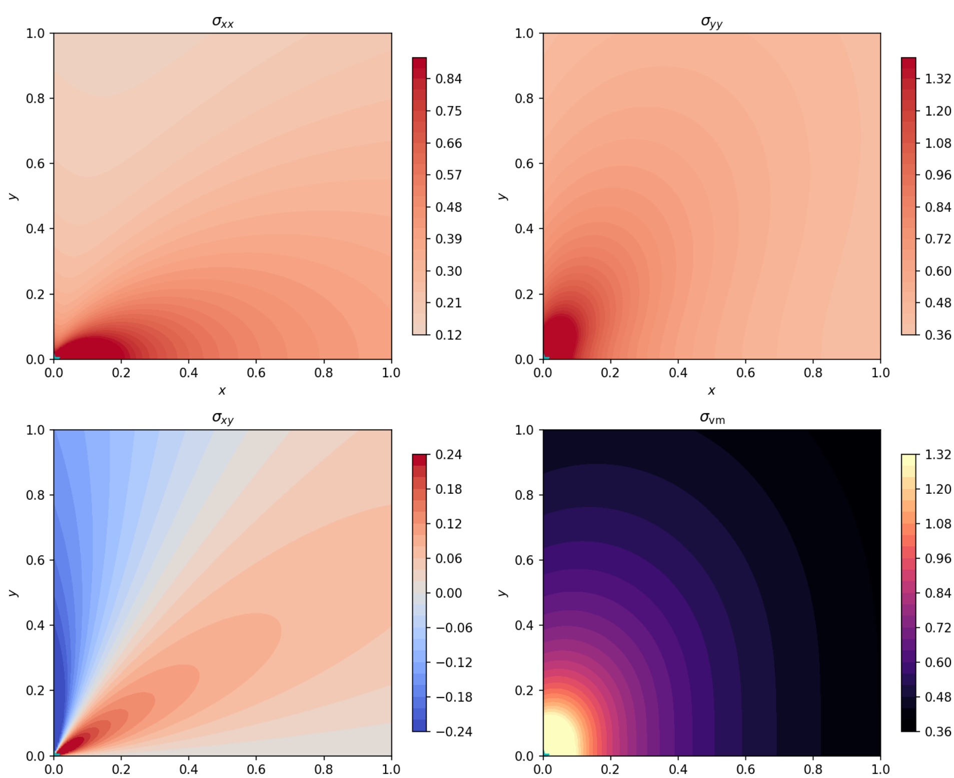


(a) Displacement field. Left: u_x; right: u_y.



(b) Stress fields. Top left: σ_{xx}; top right: σ_{yy}; bottom left: σ_{xy}; bottom right: σ_{vm} (von Mises).

Fig. 7: Mode I crack-tip solution fields at $L = 4$, $p = 16$.

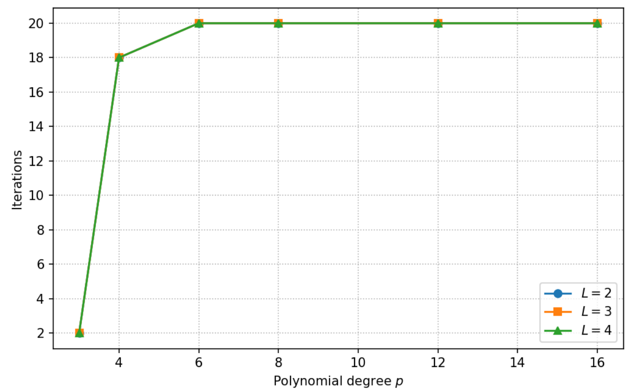
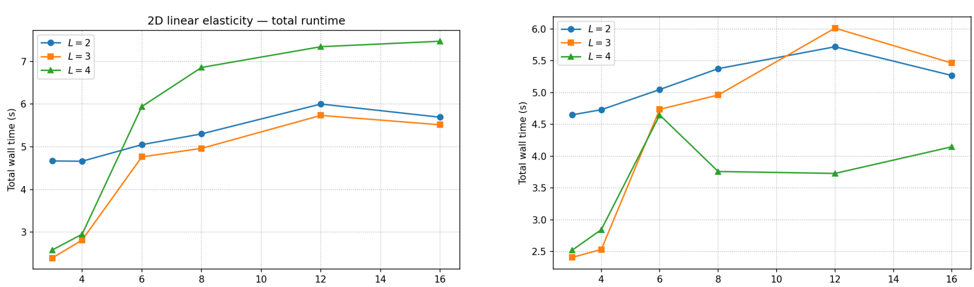


Fig. 8: ADI iteration counts for the Mode I crack-tip problem as a function of p, for each $L \in \{2, 3, 4\}$.



(a) Smooth manufactured solution.

(b) Mode I crack-tip problem.

Fig. 9: Total wall-clock time (build + solve) vs. polynomial degree p for both benchmark problems, on a CPU. Run times increase with p and L, dominated by the build stage.

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Appendix A. HPS Pseudocode.

The three algorithmic stages of the HPS method are listed below in the notation of section 4, for the 2D DtN variant. Full derivations are given in [8, 9].

Algorithm A.1 Local solve stage

Input: $\{\mathbf{L}^{(i)}\}$, $\{\mathbf{f}^{(i)}\}$, differentiation matrices.
for leaf $i = 1, \ldots, n_{\text{leaves}}$ **do**
  Apply boundary bordering; invert bordered system ▷ *cost* $\mathcal{O}(p^6)$ *per leaf*
  Extract $\mathbf{Y}^{(i)}$, $\mathbf{T}^{(i)}$, $\mathbf{v}^{(i)}$, $\mathbf{h}^{(i)}$
end for
Output: $\{\mathbf{T}^{(i)}, \mathbf{h}^{(i)}, \mathbf{Y}^{(i)}, \mathbf{v}^{(i)}\}$.

Algorithm A.2 Merge (upward pass) stage

Input: $\{\mathbf{T}^{(i)}, \mathbf{h}^{(i)}\}$ for all leaves.
for level $\ell = L - 1, \ldots, 0$ **do**
  for node j with children a, b, c, d **do**
    Assemble blocks A, B, C, D from $\{\mathbf{T}^{(a)}, \ldots, \mathbf{T}^{(d)}\}$ and child $\mathbf{h}$
    Invert D ▷ *main cost per level*
    $\mathbf{T}^{(j)} = A - BD^{-1}C$; $\mathbf{h}^{(j)} = \mathbf{h}^{(\text{ch})}_{\mathcal{E}} - BD^{-1}\mathbf{h}^{(\text{ch})}_{\mathcal{I}}$
    $\mathbf{S}^{(j)} = -D^{-1}C$; $\tilde{\mathbf{g}}^{(j)} = -D^{-1}\mathbf{h}^{(\text{ch})}_{\mathcal{I}}$
  end for
end for
Output: $\{\mathbf{S}^{(j)}, \tilde{\mathbf{g}}^{(j)}\}$ for all interior nodes.

Algorithm A.3 Downward pass (solve stage)

Input: $\mathbf{g}|_{\partial\Omega}$; $\{\mathbf{S}^{(j)}, \tilde{\mathbf{g}}^{(j)}\}$; $\{\mathbf{Y}^{(i)}, \mathbf{v}^{(i)}\}$.
$\mathbf{g}^{(\text{root})} \leftarrow \mathbf{g}|_{\partial\Omega}$
for level $\ell = 0, \ldots, L-1$ **do**
  for node j with children a, b, c, d **do**
    $\mathbf{g}_{\mathcal{I}}^{(j)} = \mathbf{S}^{(j)}\mathbf{g}_{\mathcal{E}}^{(j)} + \tilde{\mathbf{g}}^{(j)}$; distribute to children
  end for
end for
for leaf $i = 1, \ldots, n_{\text{leaves}}$ **do**
  $\mathbf{u}^{(i)} = \mathbf{Y}^{(i)}\mathbf{g}^{(i)} + \mathbf{v}^{(i)}$
end for
Output: $\{\mathbf{u}^{(i)}\}$.

Appendix B. FEM Convergence Reference.

For reference, Figure 10 shows the convergence of a standard Continuous Galerkin FEM discretization on a uniform mesh.

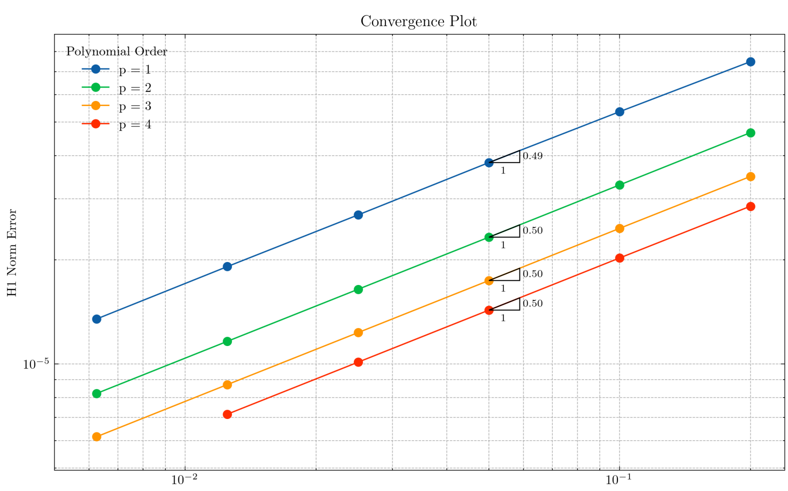

Fig. 10: Continuous Galerkin FEM convergence in the H^1 norm for the Mode I crack-tip problem. Polynomial orders $p = 1, \ldots, 4$ all yield slope ≈ 0.5 on the log-log scale, confirming the algebraic $\mathcal{O}(h^{1/2})$ rate imposed by the $\sqrt{r}$ singularity; higher p reduces the constant but not the slope.